

Assessment Tool 3: Application-Based Questions

1. Seasonal Sales Traffic Spikes in an E-Commerce Company

Application

During seasonal sales such as Black Friday or festival offers, an e-commerce website experiences sudden traffic spikes. The cloud must handle increased demand without performance affecting website.

Cloud Concepts

Auto-Scaling

- Automatically increases the number of servers when traffic increases.
- Removes extra servers when traffic decreases.
- Ensures sufficient computing resources during peak demand.

If predefined thresholds are exceeded, monitoring tools trigger auto-scaling automatically.

Load Balancing Real-World Example

During **Amazon's Great Indian Festival**, millions of customers visit simultaneously. Auto-scaling adds additional servers, while load balancers distribute traffic efficiently, ensuring uninterrupted shopping.

Conclusion

across multiple servers.

- Prevents any single server

Cloud auto-scaling, load balancing, and monitoring provide high availability, minimize downtime, improve customer experience, and optimize infrastructure costs.

2. Secure Cloud Backup Storage

Application

An organization stores large volumes of backup data in the cloud and requires secure, durable, and cost-effective storage.

Cloud Concepts

Durability

Cloud providers store multiple copies of data across different storage devices to prevent data loss.

Data Redundancy

Backup copies are replicated across multiple availability zones or regions so data remains accessible even if one location fails.

Security

Encryption

- Encrypts data before storage and during transmission.
- Prevents unauthorized access.

Access Control

Application

A mobile application is deployed worldwide and must deliver fast content with minimal delay to users in different countries.

Cloud Concepts

ContentDeliveryNetwork(CDN)

A CDN stores frequently accessed content in multiple locations around the world.

EdgeServers

Edge servers deliver content from the nearest geographical location instead of the central server.

Performance Improvement

- Reduces latency.
- Faster content delivery.
- Less network congestion.
- Better application responsiveness.

Global Availability

Cloud providers maintain data centers across multiple regions, allowing users to access services even if one region experiences failures.

Real-World Example

Streaming platforms and online multiplayer games use CDNs to deliver videos and game updates quickly to players across different countries, reducing lag and buffering.

Conclusion

CDNs and edge servers significantly improve performance, reduce latency, and provide reliable global access for mobile applications.

4.Financial Institution Security and Compliance

Application

A financial institution processes confidential customer transactions and must comply with strict regulatory requirements while protecting sensitive information.

Cloud Concepts

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- SOC 2
- GDPR (where applicable)

Logging and Monitoring

- Record all user activities.
- Detect suspicious behavior.
- Support auditing and compliance verification.

Encryption

- Encrypt data during storage and transmission.
- Protect customer financial information from unauthorized access.

Data Sovereignty

Organizations ensure sensitive customer data is stored only within approved geographic regions according to government regulations.

Real-World Example

During an online banking transaction, customer account details are encrypted, user activities are logged for auditing, and transaction data remains stored within the country's approved cloud region.

Conclusion

Cloud security features, compliance certifications, encryption, monitoring, and data sovereignty help financial institutions securely process sensitive financial transactions.

5. Containerization for Application Deployment

Application

A software development team wants faster application deployment with better scalability across multiple cloud environments.

Containers vs Virtual Machines

Containers	Virtual Machines
Share host operating system	Each VM has its own operating system
Lightweight	Heavyweight
Faster startup	Slower startup
Lower resource usage	Higher resource usage
Higher deployment speed	Slower deployment

Cloud Concepts

Kubernetes Orchestration

Kubernetes automates:

- Container deployment
- Scaling
- Load balancing

- Self-healing
- Rolling updates

Portability

Containers package the application and its dependencies together, allowing the same application to run consistently across AWS, Azure, Google Cloud, or on-premises environments.

Microservices Example

An e-commerce application can be divided into:

- User Service
- Product Service
- Payment Service
- Order Service
- Notification Service

Each service runs independently inside its own container and can be updated or scaled without affecting other services.

Conclusion

Containerization combined with Kubernetes provides fast deployment, efficient resource utilization, scalability, portability, and simplified management across cloud environments.